

Limit Aljabar dan Trigonometri

Pertemuan Ke-5

LIMIT

Untuk mengatakan $\lim_{x \to c} f(x) = L$ berarti bahwa bilamana x dekat tetapi berlainan dari c, maka $f(x)$ dekat ke L.

Limit bermakna "mendekati". Misal ada bentuk $f(x) = \frac{x^2-1}{x-1}$.

Dalam hal ini, fungsi tersebut tidak terdefinisi pada $x = 1$ karena di titik ini $f(x) = \frac{0}{0}$ yang tanpa arti. Tetapi kita masih dapat menanyakan apa yang terjadi pada x mendekati 1.

SIFAT LIMIT

1). $\lim_{x \to c} k = k$

2). $\lim_{x \to c} x = c$

3). $\lim_{x \to c} k.\, f(x) = k. \lim_{x \to c} f(x)$

4). $\lim_{x \to c} [f(x) + g(x)] = \lim_{x \to c} f(x) + \lim_{x \to c} g(x)$

5). $\lim_{x \to c} [f(x) - g(x)] = \lim_{x \to c} f(x) - \lim_{x \to c} g(x)$

6). $\lim_{x \to c} [f(x).\, g(x)] = \lim_{x \to c} f(x).\, \lim_{x \to c} g(x)$

7). $\lim_{x \to c} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x \to c} f(x)}{\lim_{x \to c} g(x)}$

dengan syarat : $\lim_{x \to c} g(x) \neq 0$

8). $\lim_{x \to c} [f(x)]^n = [\lim_{x \to c} f(x)]^n \quad , n \geq 1$

9). $\lim_{x \to c} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to c} f(x)} \quad , \lim_{x \to c} f(x) > 0$

$a.\ \lim_{x \to 4} 3x^2 - 2x$

$b.\ \lim_{x \to 4} \dfrac{\sqrt{x^2 + 9}}{x}$

LATIHAN SOAL – LIMIT ALJABAR

Carilah nilai dari limit berikut:

1. $\lim_{x \to 1} 3x + 5$
2. $\lim_{x \to 1} \frac{(x-1)^3}{x-1}$
3. $\lim_{x \to 2} \frac{x^2-7x+10}{x-2}$
4. $\lim_{x \to 1} \frac{x-1}{\sqrt{x}-1}$
5. $\lim_{x \to 4} \frac{4x-x^2}{2-\sqrt{x}}$

Sebuah benda bergerak sepanjang garis lurus dengan posisi terhadap waktu t (dalam detik) dan S dalam meter dengan fungsi:

$S(t) = \frac{t^2+4t-12}{t-2}$. Posisi benda pada saat t = 2 detik ditentukan menggunakan limit fungsi.

a. Hitunglah $\lim_{t \to 2} s(t)$!
b. Interpretasikan hasil limit tersebut!

LIMIT TRIGONOMETRI

Rumus limit fungsi trigonometri

1. $\lim_{x \to 0} \frac{\sin mx}{nx} = \frac{m}{n}$
2. $\lim_{x \to 0} \frac{mx}{\sin nx} = \frac{m}{n}$
3. $\lim_{x \to 0} \frac{\sin mx}{\sin nx} = \frac{m}{n}$
4. $\lim_{x \to 0} \frac{\tan mx}{nx} = \frac{m}{n}$
5. $\lim_{x \to 0} \frac{mx}{\tan nx} = \frac{m}{n}$
6. $\lim_{x \to 0} \frac{\tan mx}{\tan nx} = \frac{m}{n}$
7. $\lim_{x \to 0} \frac{\sin mx}{\tan nx} = \frac{m}{n}$
8. $\lim_{x \to 0} \frac{\tan mx}{\sin nx} = \frac{m}{n}$
9. $\lim_{x \to 0} \frac{\sin x}{x} = 1$
10. $\lim_{x \to 0} \frac{\tan x}{x} = 1$

TRIGONOMETRY LAWS AND IDENTITIES

TANGENT IDENTITIES

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$\cot\theta = \frac{\cos\theta}{\sin\theta}$$

RECIPROCAL IDENTITIES

$$\csc\theta = \frac{1}{\sin\theta} \qquad \sin\theta = \frac{1}{\csc\theta}$$

$$\sec\theta = \frac{1}{\cos\theta} \qquad \cos\theta = \frac{1}{\sec\theta}$$

$$\cot\theta = \frac{1}{\tan\theta} \qquad \tan\theta = \frac{1}{\cot\theta}$$

PYTHAGOREAN IDENTITIES

$$\sin^2\theta + \cos^2\theta = 1$$

$$\tan^2\theta + 1 = \sec^2\theta$$

$$\cot^2\theta + 1 = \csc^2\theta$$

PERIODIC IDENTITIES

$$\sin(\theta + 2\pi n) = \sin\theta$$

$$\cos(\theta + 2\pi n) = \cos\theta$$

$$\tan(\theta + \pi n) = \tan\theta$$

$$\csc(\theta + 2\pi n) = \csc\theta$$

$$\sec(\theta + 2\pi n) = \sec\theta$$

$$\cot(\theta + \pi n) = \cot\theta$$

EVEN/ODD IDENTITIES

$$\sin(-\theta) = -\sin\theta$$

$$\cos(-\theta) = \cos\theta$$

$$\tan(-\theta) = -\tan\theta$$

$$\csc(-\theta) = -\csc\theta$$

$$\sec(-\theta) = \sec\theta$$

$$\cot(-\theta) = -\cot\theta$$

DOUBLE ANGLE IDENTITIES

$$\sin(2\theta) = 2\sin\theta\cos\theta$$

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta$$

$$= 2\cos^2\theta - 1$$

$$= 1 - 2\sin^2\theta$$

$$\tan(2\theta) = \frac{2\tan\theta}{1 - \tan^2\theta}$$

HALF ANGLE IDENTITIES

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{2}}$$

$$\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos\theta}{2}}$$

$$\tan\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$$

LAW OF COSINES

$$a^2 = b^2 + c^2 - 2bc\cos\alpha$$

$$b^2 = a^2 + c^2 - 2ac\cos\beta$$

$$c^2 = a^2 + b^2 - 2ab\cos\gamma$$

PRODUCT TO SUM IDENTITIES

$$\sin\alpha\sin\beta = \frac{1}{2}[\cos(\alpha-\beta) - \cos(\alpha+\beta)]$$

$$\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha-\beta) + \cos(\alpha+\beta)]$$

$$\sin\alpha\cos\beta = \frac{1}{2}[\sin(\alpha+\beta) + \sin(\alpha-\beta)]$$

$$\cos\alpha\sin\beta = \frac{1}{2}[\sin(\alpha+\beta) - \sin(\alpha-\beta)]$$

SUM TO PRODUCT IDENTITIES

$$\sin\alpha + \sin\beta = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$\sin\alpha - \sin\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

$$\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$\cos\alpha - \cos\beta = -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

LAW OF SINES

$$\frac{\sin\alpha}{a} = \frac{\sin\beta}{b} = \frac{\sin\gamma}{c}$$

LAW OF TANGENTS

$$\frac{a-b}{a+b} = \frac{\tan\left[\frac{1}{2}(\alpha-\beta)\right]}{\tan\left[\frac{1}{2}(\alpha+\beta)\right]}$$

$$\frac{b-c}{b+c} = \frac{\tan\left[\frac{1}{2}(\beta-\gamma)\right]}{\tan\left[\frac{1}{2}(\beta+\gamma)\right]}$$

$$\frac{a-c}{a+c} = \frac{\tan\left[\frac{1}{2}(\alpha-\gamma)\right]}{\tan\left[\frac{1}{2}(\alpha+\gamma)\right]}$$

SUM/DIFFERENCES IDENTITIES

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$$

MOLLWEIDE'S FORMULA

$$\frac{a+b}{c} = \frac{\cos\left[\frac{1}{2}(\alpha-\beta)\right]}{\sin\left(\frac{1}{2}\gamma\right)}$$

COFUNCTION IDENTITIES

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta$$

$$\csc\left(\frac{\pi}{2} - \theta\right) = \sec\theta$$

$$\tan\left(\frac{\pi}{2} - \theta\right) = \cot\theta$$

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta$$

$$\sec\left(\frac{\pi}{2} - \theta\right) = \csc\theta$$

$$\cot\left(\frac{\pi}{2} - \theta\right) = \tan\theta$$

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α	I					II				III				IV			
	0°	30°	45°	60°	90°	120°	135°	150°	180°	210°	225°	240°	270°	300°	315°	330°	360°
	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{5\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{7\pi}{4}$	$\frac{11\pi}{6}$	2π
$\sin\alpha$	0	$\frac{1}{2}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}\sqrt{3}$	1	$\frac{1}{2}\sqrt{3}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{2}\sqrt{2}$	$-\frac{1}{2}\sqrt{3}$	-1	$-\frac{1}{2}\sqrt{3}$	$-\frac{1}{2}\sqrt{2}$	$-\frac{1}{2}$	0
$\cos\alpha$	1	$\frac{1}{2}\sqrt{3}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{2}\sqrt{2}$	$-\frac{1}{2}\sqrt{3}$	-1	$-\frac{1}{2}\sqrt{3}$	$-\frac{1}{2}\sqrt{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}\sqrt{3}$	1
$\tan\alpha$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	td	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	td	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0
$\csc\alpha$	td	2	$\sqrt{2}$	$\frac{2}{3}\sqrt{3}$	1	$\frac{2}{3}\sqrt{3}$	$\sqrt{2}$	2	td	-2	$-\sqrt{2}$	$-\frac{2}{3}\sqrt{3}$	-1	$-\frac{2}{3}\sqrt{3}$	$-\sqrt{2}$	-2	td
$\sec\alpha$	1	$\frac{2}{3}\sqrt{3}$	$\sqrt{2}$	2	td	-2	$-\sqrt{2}$	$-\frac{2}{3}\sqrt{3}$	-1	$-\frac{2}{3}\sqrt{3}$	$-\sqrt{2}$	-2	td	2	$\sqrt{2}$	$\frac{2}{3}\sqrt{3}$	-1
$\cot\alpha$	td	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0	$-\frac{1}{\sqrt{3}}$	-1	$-\sqrt{3}$	td	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	1	$-\frac{1}{\sqrt{3}}$	-1	$-\sqrt{3}$	td

LATIHAN SOAL – LIMIT TRIGONMETRI

Carilah nilai dari limit berikut:

1. $\lim_{x \to 0} \frac{2 \sin 4x}{7 \tan 7x}$
2. $\lim_{x \to \frac{\pi}{2}} (\sin x + \cos x)$

LATIHAN SOAL – LIMIT TRIGONMETRI

Carilah nilai dari limit berikut:

3. $\lim_{x \to 0} \frac{2x^2 + x}{\sin x}$

4. $\lim_{x \to 0} \frac{\tan 2x \cdot \tan 3x}{3x^2}$

5. $\lim_{x \to 0} \frac{1 - \cos x}{\sin x}$

LATIHAN SOAL – LIMIT TRIGONMETRI

Carilah nilai dari limit berikut:

6. $\lim_{x \to 0} \frac{\cos 2x}{\sin x - \cos x}$

7. $\lim_{x \to \frac{\pi}{6}} \frac{1}{2} \cos 12x$

8. $\lim_{x \to 0} \frac{2x \sin 3x}{\tan^2 5x}$

9. $\lim_{x \to 0} \frac{1 - \cos 2x}{x^2}$

Thank you

